Area 2: Mechanics of Materials

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

Contents

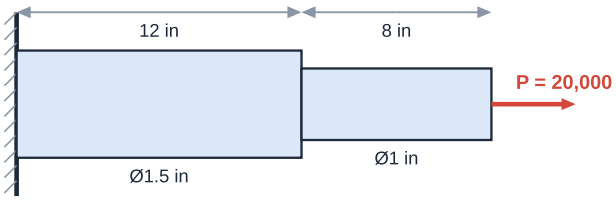
★ 2.A Axial Loading
★ 2.B Shear and Transverse Loading
★ 2.C Bending
★ 2.D Buckling
★ 2.E Torsion
★ 2.F Combined Loading
★ 2.G Static Failure
★ 2.H Fatigue Failure
★ 2.I Thermal and Interference Stresses

2.A Axial Loading

1. Key concepts

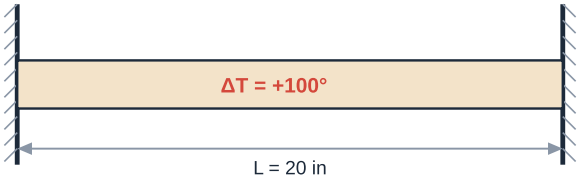
- Normal stress σ = P/A acts on a section perpendicular to the load. Elongation δ = PL/(AE) for each uniform segment; add the segments of a stepped bar.
- Statically indeterminate bars: write equilibrium, then compatibility (equal deformations, or deformations that sum to a gap), then the force–deformation relations.
- Parallel members sharing a rigid plate split the load in proportion to their axial stiffness k = AE/L.
- Poisson effect: lateral strain = −ν × axial strain. Diameter change Δd = −ν·ε·d.

Diagrams



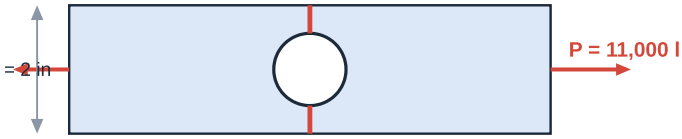
δ = Σ PL/(AE)

Stepped bar: add the δ of each segment.



σ = EαΔT (fully restrained)

Bar between rigid walls.



net area = (w − d)·t; hole Ø0.75, t = 0.5 in

Plate with a hole: net section carries the load.

2. Essential formulas

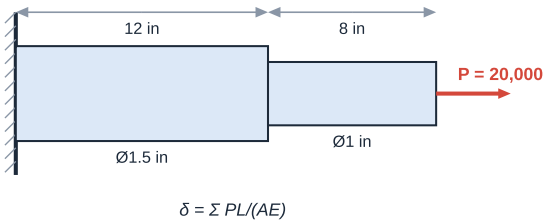
Name	Formula	Variables / units
Axial stress	$\sigma = P/A$	psi, MPa
Elongation	$\delta = PL/(AE)$; $\delta_total = \Sigma\, P_i L_i/(A_i E_i)$	in, mm
Axial stiffness	$k = AE/L$	
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1{+}k_2)$	equal strain
Poisson	$\Delta d = -\nu(\sigma/E)d$	

3. Handbook search keywords

axial deformation PL/AE | statically indeterminate | compatibility | Poisson's ratio

4. Worked examples

Example 1: Stepped bar
A steel bar ($E = 30\times10^6$ psi) has 12 in of Ø1.5 in and 8 in of Ø1.0 in and carries 20,000 lb tension. Find the total elongation.
Handbook search: **axial deformation**

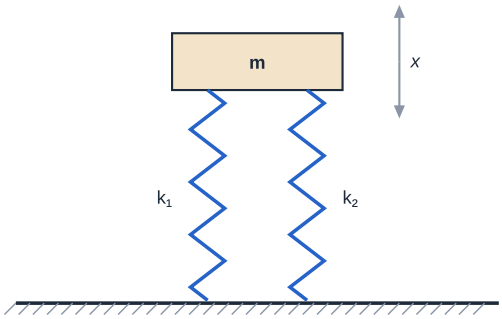


1. $A_1 = 1.7671$ in², $A_2 = 0.7854$ in².
2. $\delta = (20{,}000/30\times10^6)(12/1.7671 + 8/0.7854) = 0.01132$ in.

Answer: 0.0113 in

Check your answer: The thin segment (smaller A) should contribute most of the stretch.

Example 2: Steel and aluminum posts
A rigid cap sits on a steel post ($A = 1.0$ in², $E = 30\times10^6$) and an aluminum post ($A = 1.5$ in², $E = 10\times10^6$) of equal length and carries 30,000 lb. Find the steel load.
Handbook search: **statically indeterminate**

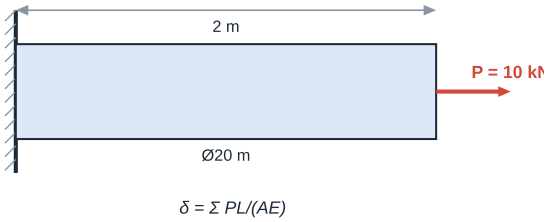


1. Compatibility: equal shortening, so the load splits in proportion to AE.
2. P_s = 30,000·(30)/(30 + 15) = 20,000 lb; P_a = 10,000 lb.

Answer: 20,000 lb

Check your answer: Loads in the two posts must add up to the applied load, and the stiffer post (larger AE/L) takes more.

Try it yourself: Elongation in SI
A steel rod ($E = 200$ GPa), Ø20 mm and 2.0 m long, carries 10 kN. Find δ.
Handbook search: **axial deformation**

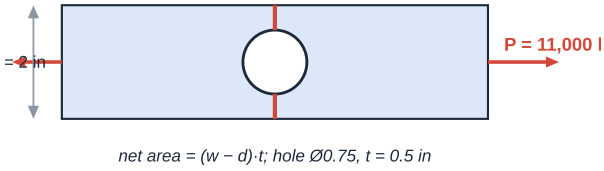


1. $A = \pi(20)^2/4 = 314.2$ mm².
2. $\delta = PL/AE = 10{,}000 \times 2{,}000/(314.2 \times 200{,}000) = 0.318$ mm.

Answer: 0.318 mm

Check your answer: Keep N, mm and MPa together (MPa = N/mm²). A result in metres usually means mixed units.

Example 4: Net-section stress
A 2 in × 0.5 in plate has one 0.75 in hole and carries 11,000 lb. Find the average net-section stress.
Handbook search: **net section**



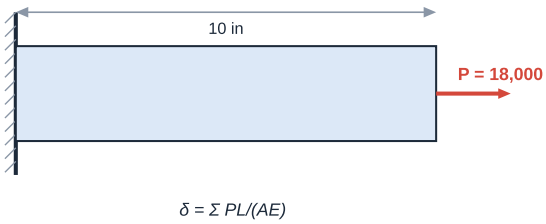
1. $A_{net} = (2 - 0.75) \times 0.5 = 0.625$ in².
2. $\sigma = 11{,}000/0.625 = 17{,}600$ psi.

Answer: 17,600 psi

Check your answer: Net-section stress must be higher than the gross value 11,000/1.0 = 11,000 psi.

Example 5: Factor of safety of a link
A tension link (A = 1.25 in², Sy = 36,000 psi) carries 18,000 lb. Find n against yield.

Handbook search: **factor of safety**



1. σ = 18,000/1.25 = 14,400 psi.
2. n = Sy/σ = 36,000/14,400 = 2.50.

Answer: n = 2.50

Check your answer: n above 1 means it does not yield. Equivalently, the yield load 45,000 lb is 2.5 × the applied load.

5. Common traps

- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.A Axial Loading: one-page cheat summary

► $\sigma = P/A$
► $\delta = \Sigma PL/AE$
► $k = AE/L$
► Parallel: $P \propto AE/L$
► $\Delta d = -\nu(\sigma/E)d$

Formulas

Axial stress	$\sigma = P/A$
Elongation	$\delta = PL/(AE)$; $\delta_total = \Sigma\ P_iL_i/(A_iE_i)$
Axial stiffness	$k = AE/L$
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1+k_2)$
Poisson	$\Delta d = -\nu(\sigma/E)d$

Search keywords

axial deformation PL/AE, statically indeterminate, compatibility, Poisson's ratio

Top traps

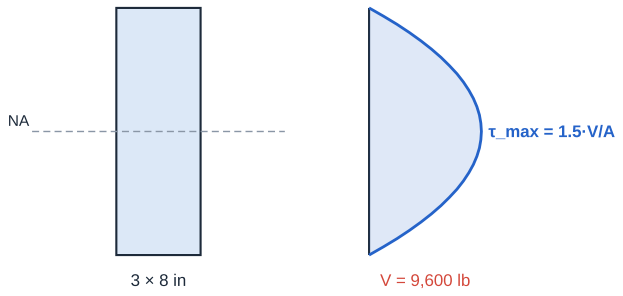
- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.B Shear and Transverse Loading

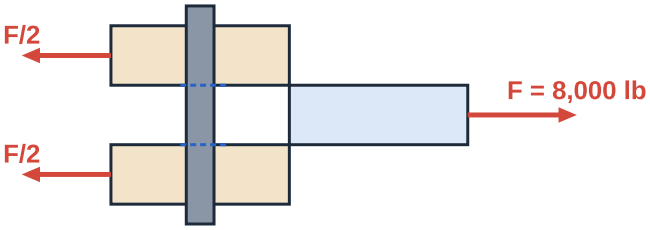
1. Key concepts

- Direct shear (pins, bolts, keys): τ = V/A_shear. Single shear uses 1 area per pin; double shear uses 2.
- Beam transverse shear: τ = VQ/(Ib). It peaks at the neutral axis and is zero at the top and bottom fibers.
- Rectangle: τ_max = 1.5V/A. Solid circle: τ_max = (4/3)V/A. Thin-walled tube: τ_max ≈ 2V/A.
- Shear flow q = VQ/I (force per length) sizes the nails, bolts or welds that join built-up beam parts.

Diagrams

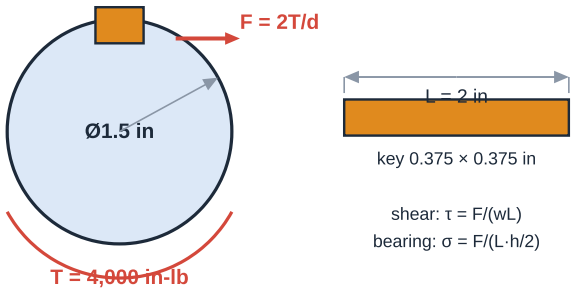


Parabolic shear distribution in a rectangle.



Ø0.625 in pin, double shear: τ = F/(2A)

Pin in double shear: two shear planes.



Key in shear on the shaft surface.

2. Essential formulas

Name	Formula	Variables / units
Direct shear	$\tau = V/(nA)$	n = number of shear planes
Transverse shear	$\tau = VQ/(Ib)$	Q = first moment of the area beyond the cut
Rectangle max	$\tau_{max} = 3V/(2A)$	
Circle max	$\tau_{max} = 4V/(3A)$	
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$	lb/in

3. Handbook search keywords

transverse shear stress VQ/Ib · shear flow · double shear · first moment of area

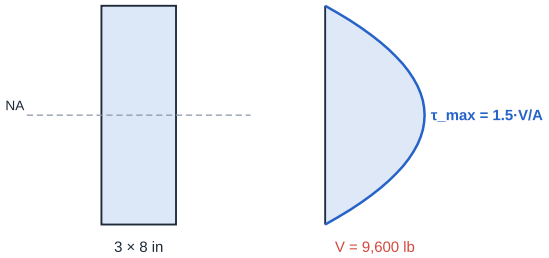
4. Worked examples

Example 1: Rectangular beam
A 3 in × 8 in rectangular beam carries V = 9,600 lb. Find τ_max.
Handbook search: **transverse shear**

1. A = 24 in².
2. τ_max = 1.5(9,600)/24 = 600 psi at the neutral axis.

Answer: 600 psi

Check your answer: τ_max is 1.5 × the average shear V/A (= 400 psi here).

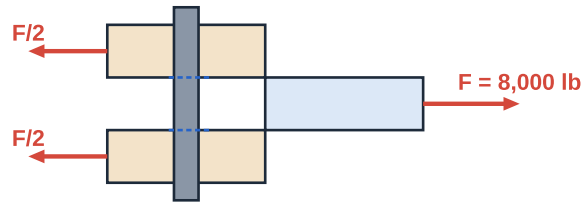


Example 2: Clevis pin in double shear
A Ø0.625 in pin carries 8,000 lb in double shear. Find τ.
Handbook search: **double shear**

1. A = π(0.625)²/4 = 0.3068 in².
2. τ = 8,000/(2 × 0.3068) = 13,038 psi.

Answer: 13,038 psi

Check your answer: Double shear halves the stress compared with single shear.



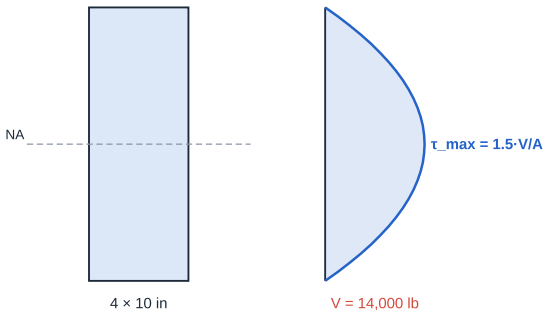
Ø0.625 in pin, double shear: τ = F/(2A)

Try it yourself: Max shear in a rectangle
A 4 in × 10 in beam carries V = 14,000 lb. Find τ_max.
Handbook search: **transverse shear**

1. τ_avg = V/A = 14,000/40 = 350 psi.
2. τ_max = 1.5 × τ_avg = 525 psi.

Answer: 525 psi

Check your answer: Peak is at the neutral axis and zero at the top and bottom faces.

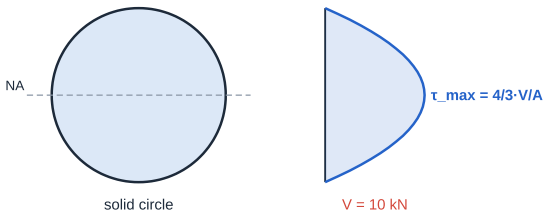


Example 4: Solid circular section
A Ø40 mm round pin used as a beam carries V = 10 kN. Find τ_max.
Handbook search: **transverse shear circular**

1. A = π(40)²/4 = 1257 mm².
2. τ_max = (4/3)V/A = 10.61 MPa.

Answer: 10.61 MPa

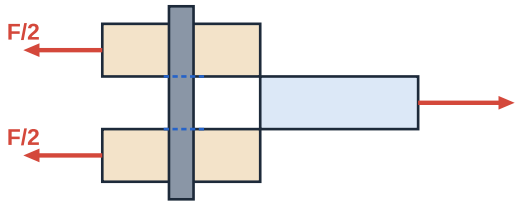
Check your answer: For a circle the factor is 4/3, not 1.5.



Example 5: Shear pin through a shaft
A Ø0.375 in pin through a Ø1.0 in shaft (double shear) transmits 5,000 in-lb. Find τ.

Handbook search: **shear pin**

1. The two shear planes sit at the shaft surface, d = 1.0 in apart, so their forces form a couple: T = F·d → F = 5,000/1.0 = 5,000 lb on each plane.
2. Area per plane = π(0.375)²/4 = 0.1104 in².
3. τ = F/A = 5,000/0.1104 = 45,271 psi.



Ø0.375 in pin, double shear: τ = F/(2A)

Answer: 45,271 psi

Check your answer: Back-check: τ·A·d = T, so the stress times 0.1104 in² times 1.0 in should return 5,000 in-lb.

5. Common traps

- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.B Shear and Transverse Loading: one-page cheat summary

- ► $\tau = V/(nA)$
- ► $\tau = VQ/Ib$
- ► Rect $1.5V/A$; circle $4V/3A$; tube $2V/A$
- ► $q = VQ/I$

Formulas

Direct shear	$\tau = V/(nA)$
Transverse shear	$\tau = VQ/(Ib)$
Rectangle max	$\tau_{max} = 3V/(2A)$
Circle max	$\tau_{max} = 4V/(3A)$
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$

Search keywords

transverse shear stress VQ/Ib, shear flow, double shear, first moment of area

Top traps

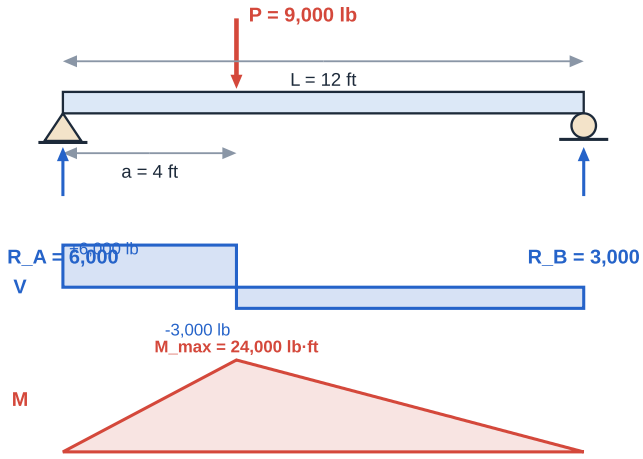
- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.C Bending

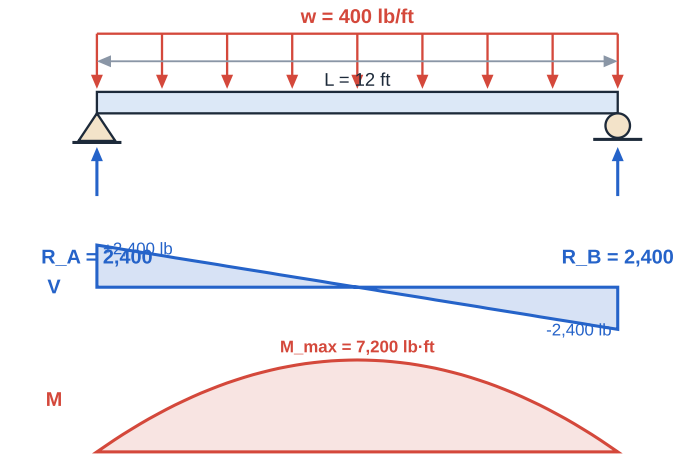
1. Key concepts

- Draw shear and moment diagrams. Shear is the slope of the moment curve, so moment peaks where V = 0. Jumps happen at point loads (V) and applied couples (M).
- Flexure formula: σ = Mc/I = M/S, with tension and compression on opposite sides of the neutral axis.
- Parallel axis: I = I_c + Ad². For a composite section, find the centroid first.
- Deflection: use the handbook beam tables and superpose load cases. Simply supported with a center load: PL³/48EI. Cantilever with an end load: PL³/3EI. UDL on a simple span: 5wL⁴/384EI.

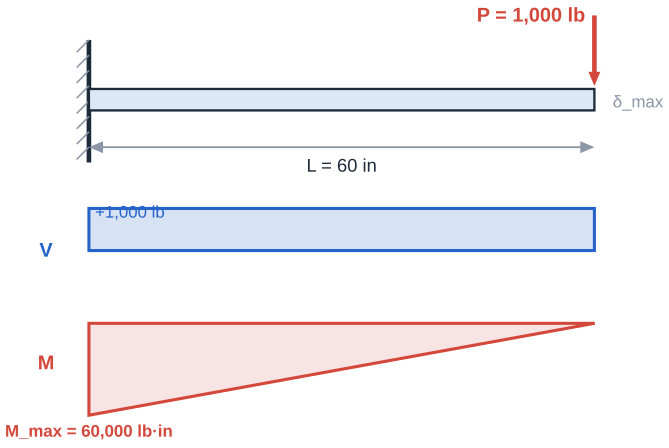
Diagrams



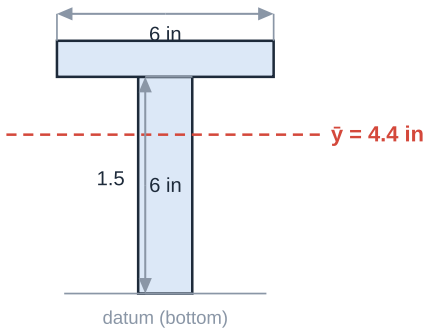
Simply supported beam with a point load: shear and moment diagrams.



Uniform load: parabolic moment, M_max = wL²/8.



Cantilever: maximum moment at the wall.



Composite section centroid.

2. Essential formulas

Name	Formula	Variables / units
Flexure	$\sigma = Mc/I = M/S$	S = I/c
Rectangle	$I = bh^3/12\ ;\ S = bh^2/6$	
Circle	$I = \pi d^4/64\ ;\ S = \pi d^3/32$	
Parallel axis	$I = I_c + Ad^2$	

SS center load	$M_{max} = PL/4 \; ; \; \delta = PL^3/48EI$	
SS UDL	$M_{max} = wL^2/8 \; ; \; \delta = 5wL^4/384EI$	
Cantilever end load	$M_{max} = PL \; ; \; \delta = PL^3/3EI$	
Cantilever UDL	$M_{max} = wL^2/2 \; ; \; \delta = wL^4/8EI$	

3. Handbook search keywords

flexure formula Mc/I | shear and moment diagrams | beam deflection tables | section modulus | parallel axis theorem

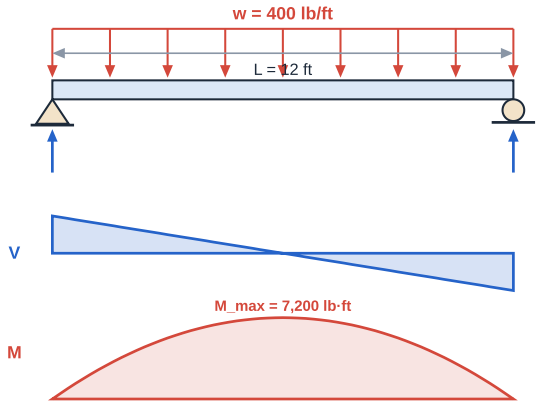
4. Worked examples

Example 1: Wood beam with UDL
A 12 ft simple span carries 400 lb/ft. The section is 3.5 in × 9.25 in. Find σ_max.
Handbook search: **bending stress**

1. M = wL²/8 = 400(12)²/8 = 7,200 ft-lb = 86,400 in-lb.
2. S = bh²/6 = 49.9 in³.
3. σ = M/S = 1,731 psi.

Answer: 1,731 psi

Check your answer: M_max = wL²/8 = 7,200 lb·ft. Remember to convert to lb·in (× 12) before dividing by S.

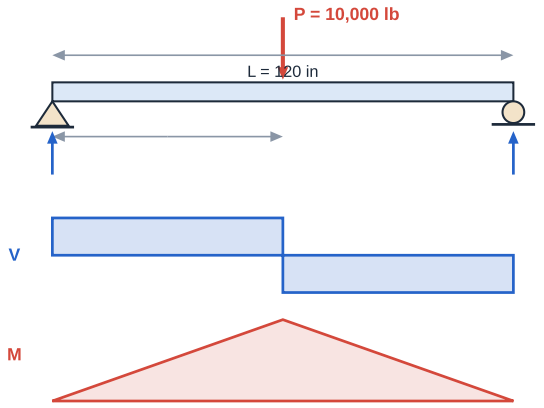


Example 2: Midspan deflection
A simply supported steel beam (E = 29×10⁶ psi, I = 100 in⁴) spans 120 in with 10,000 lb at midspan. Find δ.
Handbook search: **beam deflection**

1. δ = PL³/48EI = 10,000(120)³/(48·29×10⁶·100) = 0.1241 in.

Answer: 0.1241 in

Check your answer: Deflection of a stiff steel beam is a small fraction of an inch; L/δ should be in the hundreds.

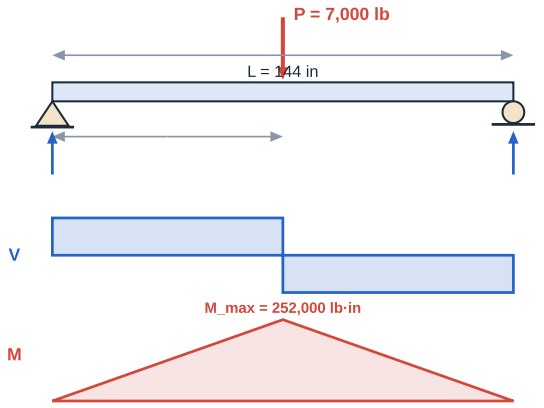


Try it yourself: Point load at midspan
A simply supported 4 in × 8 in beam spans 144 in with 7,000 lb at midspan. Find σ_max.
Handbook search: **flexure formula**

1. M = PL/4 = 7,000 × 144/4 = 252,000 in-lb.
2. S = bh²/6 = 4 × 8²/6 = 42.67 in³.
3. σ = M/S = 5,906 psi.

Answer: 5,906 psi

Check your answer: Compare with the shear diagram: R_A = 3,500 lb, and M_max = 3,500 × 72 = 252,000 in-lb.

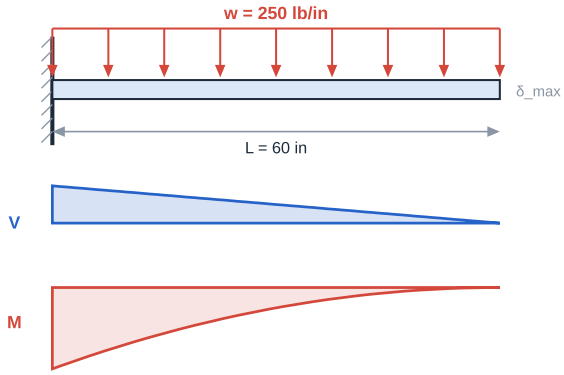


Example 4: Cantilever with a UDL
An aluminum cantilever (E = 10×10⁶ psi, I = 4 in⁴), 60 in long, carries 250 lb/in. Find the tip deflection.
Handbook search: **cantilever deflection**

1. δ = wL⁴/(8EI) = 250 × 60⁴/(8 × 10×10⁶ × 4) = 10.125 in.

Answer: 10.125 in

Check your answer: Units: lb/in × in⁴ / (psi × in⁴) = in. A big deflection here is plausible because the load is very large.

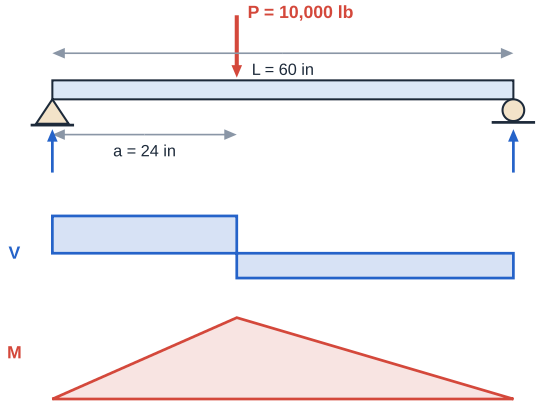


Example 5: Off-center load deflection
A simply supported beam (E = 29×10⁶ psi, I = 20 in⁴, L = 60 in) has 10,000 lb at 24 in from the left. Find δ_max.
Handbook search: **beam deflection**

1. b = 60 − 24 = 36 in (distance from the load to the far support).
2. δ_max occurs at x = √((L² − b²)/3) = 27.7 in.
3. δ_max = Pb(L² − b²)^1.5/(9√3·EIL) = 0.0734 in.

Answer: 0.0734 in

Check your answer: It must be a bit less than the midspan-load value PL³/48EI = 0.0776 in.



5. Common traps

- Leaving the moment in ft-lb while S is in in³ (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).
- Mixing the cantilever and simply supported formulas.

2.C Bending: one-page cheat summary

- ► $\sigma = Mc/I = M/S$
- ► Rect $S = bh^2/6$
- ► Max M where V = 0
- ► SS: $PL/4, wL^2/8$
- ► δ: $PL^3/48EI, 5wL^4/384EI, PL^3/3EI, wL^4/8EI$
- ► $I = I_c + Ad^2$

Formulas

Flexure	$\sigma = Mc/I = M/S$
Rectangle	$I = bh^3/12$; $S = bh^2/6$
Circle	$I = \pi d^4/64$; $S = \pi d^3/32$
Parallel axis	$I = I_c + Ad^2$
SS center load	$M_{max} = PL/4$; $\delta = PL^3/48EI$
SS UDL	$M_{max} = wL^2/8$; $\delta = 5wL^4/384EI$
Cantilever end load	$M_{max} = PL$; $\delta = PL^3/3EI$
Cantilever UDL	$M_{max} = wL^2/2$; $\delta = wL^4/8EI$

Search keywords

flexure formula Mc/I, shear and moment diagrams, beam deflection tables, section modulus, parallel axis theorem

Top traps

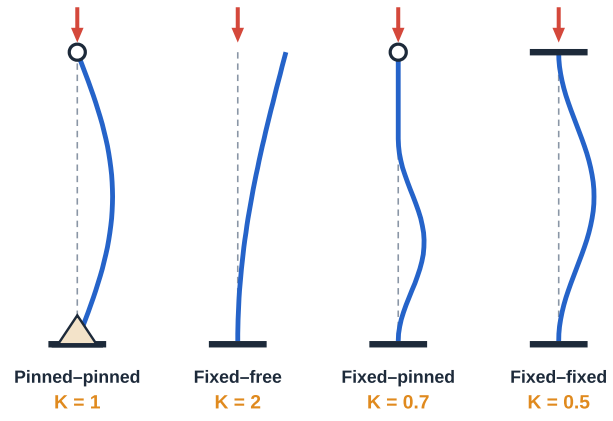
- Leaving the moment in ft-lb while S is in in^3 (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).

2.D Buckling

1. Key concepts

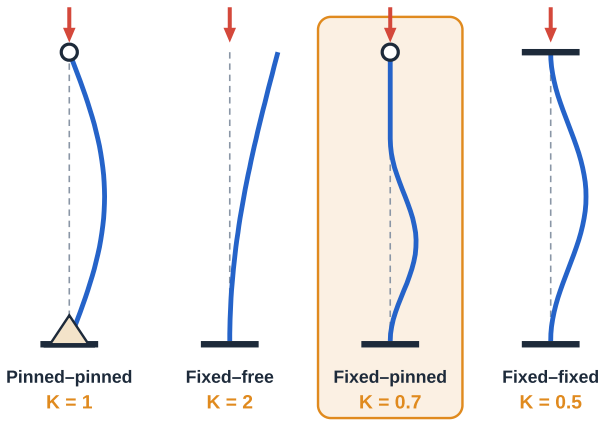
- Long slender columns fail by elastic buckling (Euler). Short columns fail by yielding. Intermediate columns follow the Johnson parabola.
- Effective length L_e = KL. Theoretical K: fixed–fixed 0.5, fixed–pinned 0.7, pinned–pinned 1.0, fixed–free 2.0. Design (recommended) K values are a little higher.
- Slenderness ratio = KL/r, with r = √(I/A). Compare it with the transition value C_c = √(2π²E/Sy). Above C_c use Euler; below it use Johnson.
- Always buckle about the weak axis (smallest I).

Diagrams



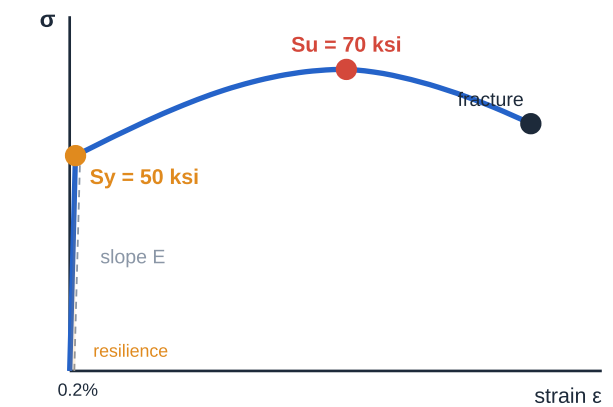
P_cr = π²EI/(KL)²: smaller K → higher buckling load

Effective-length factor K for the four classic end conditions.



P_cr = π²EI/(KL)²: smaller K → higher buckling load

Fixed–pinned column (K = 0.7).



Short columns fail by yielding; long ones buckle elastically.

2. Essential formulas

Name	Formula	Variables / units
Euler load	$P_{cr} = \pi^2 EI/(KL)^2$	
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$	
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$	
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$	

Johnson	$P_{cr} = A[S_y - (S_y \cdot KL/r)^2/(4\pi^2 E)]$	intermediate

3. Handbook search keywords

Euler buckling | effective length factor | slenderness ratio | Johnson formula | radius of gyration

4. Worked examples

Example 1: Pinned steel column

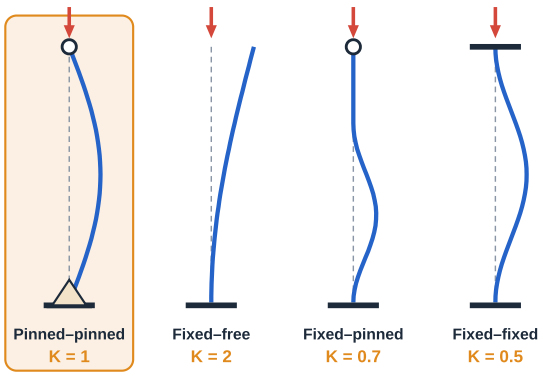
A Ø1.25 in solid steel rod (E = 29×10⁶ psi, Sy = 50 ksi), 60 in long and pinned at both ends. Find P_{cr}.

Handbook search: **Euler buckling**

1. r = d/4 = 0.3125 in; KL/r = 60/0.3125 = 192.
2. $C_c = \sqrt{(2\pi^2 \cdot 29\times10^6/50{,}000)} = 107$. Since 192 > 107, use Euler.
3. $I = \pi d^4/64 = 0.1198$ in⁴; $P_{cr} = \pi^2 EI/L^2 = 9{,}528$ lb.

Answer: 9,528 lb

Check your answer: Compare the slenderness ratio with C_c. Above C_c use Euler; below it use Johnson.



$P_{cr} = \pi^2 EI/(KL)^2$: smaller K → higher buckling load

Try it yourself: Fixed–fixed Euler load

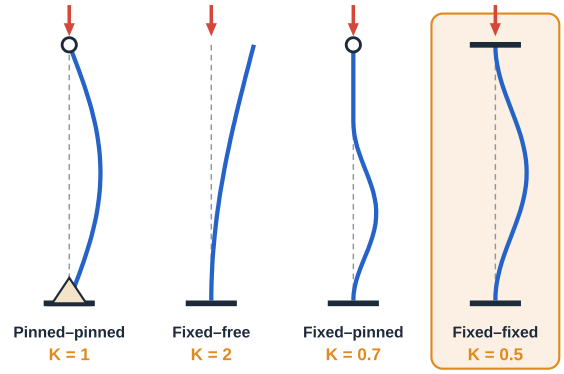
A Ø1.0 in steel column (E = 29×10⁶ psi), 48 in long, fixed–fixed. Find P_{cr}.

Handbook search: **effective length factor**

1. $I = \pi d^4/64 = 0.04909$ in⁴; K = 0.5 → KL = 24 in.
2. $P_{cr} = \pi^2 EI/(KL)^2 = 24{,}392$ lb.
3. Check slenderness: r = d/4 = 0.25 in, KL/r = 96, which is long enough for Euler with typical Sy.

Answer: 24,392 lb

Check your answer: Fixed–fixed has 4× the pinned–pinned load. The pinned value is 6,098 lb.



$P_{cr} = \pi^2 EI/(KL)^2$: smaller K → higher buckling load

Example 3: Fixed–pinned column

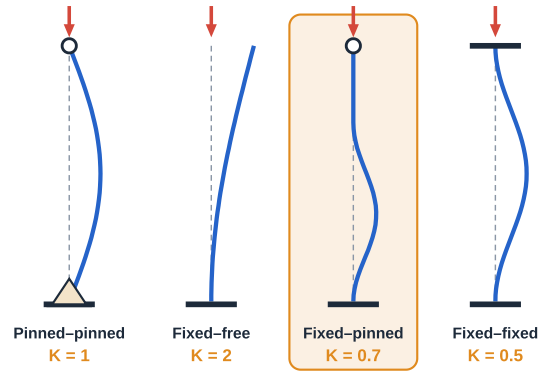
A Ø2.0 in steel column, 72 in long, fixed–pinned (E = 29×10⁶ psi). Find P_{cr}.

Handbook search: **effective length factor**

1. $I = \pi(2)^4/64 = 0.7854$ in⁴; KL = 0.7 × 72 = 50.4 in.
2. $P_{cr} = \pi^2 EI/(KL)^2 = 88{,}497$ lb.

Answer: 88,497 lb

Check your answer: K = 0.7 means about 2× the pinned–pinned load ($1/0.7^2 \approx 2.04$).



$P_{cr} = \pi^2 EI/(KL)^2$: smaller K → higher buckling load

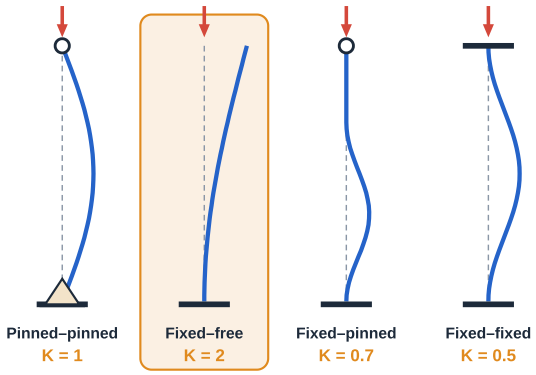
Example 4: Johnson (intermediate) column
Ø2.0 in steel, Sy = 50 ksi, E = 29×10⁶ psi, L = 20 in, K = 2.0. Find P_cr.

Handbook search: **Johnson formula**

1. r = d/4 = 0.5 in; KL/r = 80.
2. C_c = √(2π²E/Sy) = 107.0. KL/r < C_c, so use Johnson.
3. P_cr = A[Sy − (Sy·KL/r/2π)²/E] = 113,175 lb.

Answer: 113,175 lb

Check your answer: Johnson must give less than the squash load A·Sy = 157,080 lb.



P_cr = π²EI/(KL)²: smaller K → higher buckling load

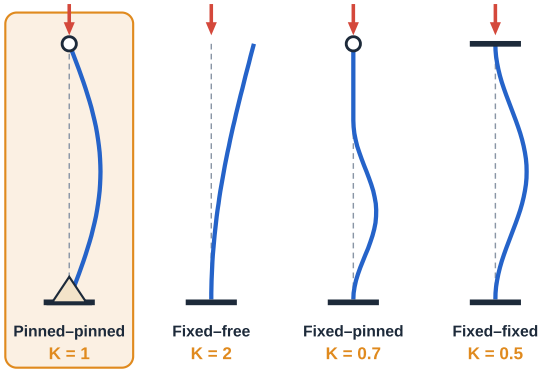
Example 5: Slenderness check first
Ø1.25 in rod, Sy = 50 ksi, E = 29×10⁶ psi, 60 in, pinned–pinned. Euler or Johnson? Find P_cr.

Handbook search: **slenderness ratio**

1. KL/r = 60/0.3125 = 192; C_c = 107.0.
2. KL/r > C_c, so Euler applies.
3. P_cr = π²EI/L² = 9,528 lb.

Answer: 9,528 lb (Euler)

Check your answer: The Euler stress P/A = 7,764 psi is below Sy/2, which confirms the elastic range.



P_cr = π²EI/(KL)²: smaller K → higher buckling load

5. Common traps

- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.D Buckling: one-page cheat summary

- ► $P_{cr} = \pi^2 EI/(KL)^2$
- ► K: 0.5 / 0.7 / 1 / 2
- ► $r = \sqrt{(I/A)}$, round $d/4$
- ► $C_c = \sqrt{(2\pi^2 E/Sy)}$
- ► Johnson below C_c

Formulas

Euler load	$P_{cr} = \pi^2 EI/(KL)^2$
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$
Johnson	$P_{cr} = A[Sy - (Sy \cdot KL/r)^2/(4\pi^2 E)]$

Search keywords

Euler buckling, effective length factor, slenderness ratio, Johnson formula, radius of gyration

Top traps

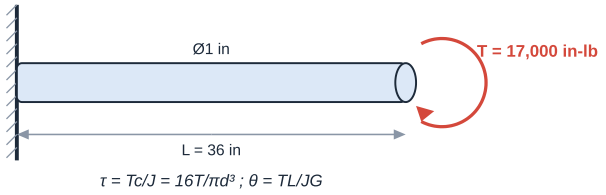
- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.E Torsion

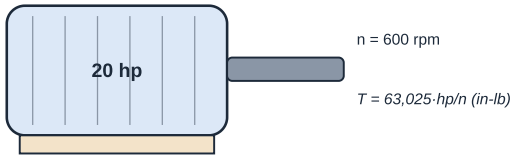
1. Key concepts

- Circular shafts: τ = Tc/J peaks at the outer surface. Angle of twist θ = TL/(JG) in radians.
- Power and torque: T(in-lb) = 63,025·hp/rpm; T(ft-lb) = 5,252·hp/rpm; T(N·m) = 9,549·kW/rpm.
- Hollow shafts: J = π(D⁴ − d⁴)/32. For the same weight they carry more torque.
- Thin-walled closed tube: τ = T/(2A_m t), where A_m is the area enclosed by the mid-wall line.

Diagrams



Solid shaft in torsion.



Power, torque and speed.

2. Essential formulas

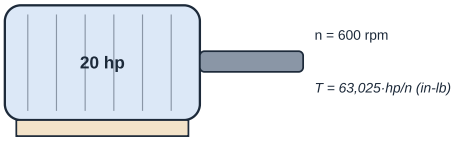
Name	Formula	Variables / units
Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)	
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4-d^4)/32$	
Twist	$\theta = TL/(JG)$ [rad]	G steel ≈ 11.5×10⁶ psi = 80 GPa
Power	$T = 63{,}025\ hp/n$ (in-lb)	
Thin tube	$\tau = T/(2A_m\ t)$	

3. Handbook search keywords

torsion shear stress Tc/J | angle of twist | polar moment of inertia | power torque speed

4. Worked examples

Example 1: Size a shaft for torque
A shaft transmits 20 hp at 600 rpm with an allowable τ of 12,000 psi. Find the minimum solid diameter.
Handbook search: **power torque speed**

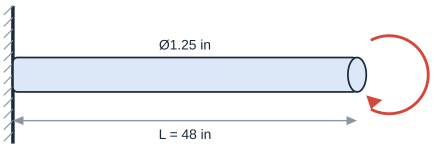


1. T = 63,025(20)/600 = 2,101 in-lb.
2. d = (16T/πτ)^(1/3) = 0.962 in.

Answer: 0.962 in (use the next standard size)

Check your answer: T = 63,025 × 20/600 = 2,101 in-lb. A slower shaft needs more torque for the same power.

Example 2: Twist
Same torque on a Ø1.25 in shaft, 48 in long, with G = 11.5×10⁶ psi. Find θ.
Handbook search: **angle of twist**

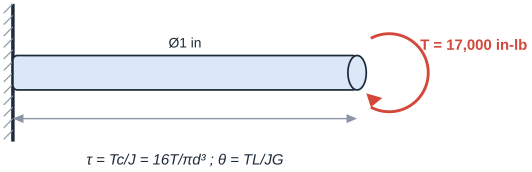


1. J = π(1.25)⁴/32 = 0.2397 in⁴.
2. θ = TL/JG = 0.0366 rad = 2.10°.

Answer: 2.10°

Check your answer: θ in radians first; multiply by 180/π for degrees.

Try it yourself: Torsional shear
A Ø1.0 in shaft carries 17,000 in-lb. Find τ_max.
Handbook search: **torsion**

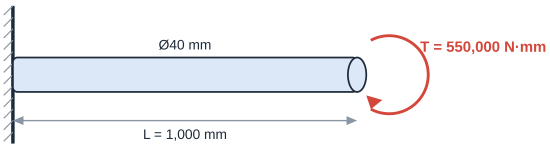


1. τ = 16T/(πd³) = 16 × 17,000/π = 86,580 psi.

Answer: 86,580 psi

Check your answer: Doubling d cuts τ by 8 (d³).

Example 4: Twist in SI
A Ø40 mm shaft (G = 80 GPa), 1 m long, carries 550 N·m. Find θ in degrees.
Handbook search: **angle of twist**

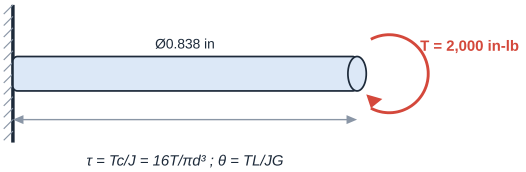


1. J = πd⁴/32 = 251,327 mm⁴.
2. θ = TL/(JG) = 550,000 × 1,000/(J × 80,000) = 0.00003 rad = 0.002°.

Answer: 0.002°

Check your answer: Use N·mm, mm and MPa together.

Example 5: Static shaft sizing
T = 2,000 in-lb, Sy = 60 ksi, Ssy = 0.577Sy, n = 2. Find d_min.
Handbook search: **shaft design**



1. τ_allow = 0.577 × 60,000/2 = 17,310 psi.
2. d = (16T/(π τ_allow))^(1/3) = 0.838 in.

Answer: 0.838 in

Check your answer: Put d back in: 16T/πd³ should return about 17,310 psi.

5. Common traps

- Leaving θ in radians when degrees are asked.
- Using I (πd⁴/64) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.E Torsion: one-page cheat summary

- ► $\tau = 16T/\pi d^3$
- ► $J = \pi d^4/32$
- ► $\theta = TL/JG$ (rad)
- ► $T = 63{,}025\ hp/n$ in-lb
- ► Thin tube $\tau = T/2A_m t$

Formulas

Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4 - d^4)/32$
Twist	$\theta = TL/(JG)$ [rad]
Power	$T = 63{,}025\ hp/n$ (in-lb)
Thin tube	$\tau = T/(2A_m t)$

Search keywords

torsion shear stress Tc/J, angle of twist, polar moment of inertia, power torque speed

Top traps

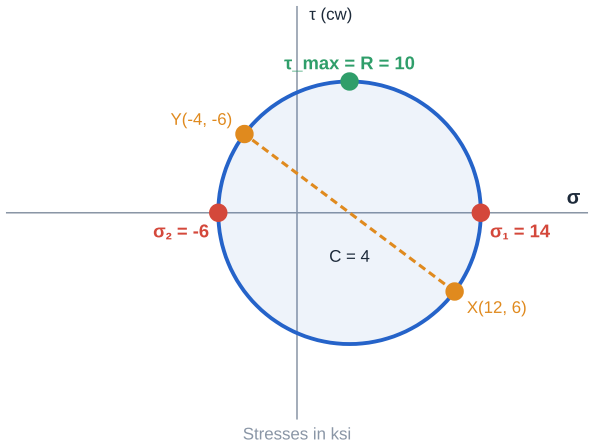
- Leaving θ in radians when degrees are asked.
- Using I ($\pi d^4/64$) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.F Combined Loading

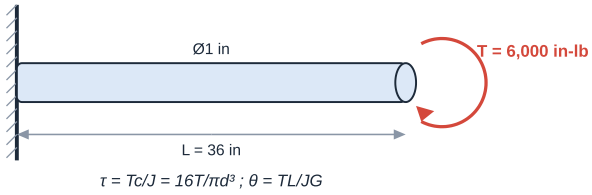
1. Key concepts

- Superpose stresses at one point: axial P/A, bending ±Mc/I, torsion Tc/J, transverse VQ/Ib. Find the critical point (usually the outer fiber).
- Mohr's circle: center C = (σx+σy)/2, radius R = √[((σx−σy)/2)² + τ²]. $\sigma_{1,2}$ = C ± R and τ_max,in-plane = R.
- In 3-D, the absolute maximum shear = (σ_max − σ_min)/2, counting σ_3 = 0 in plane stress.
- Shaft with bending plus torsion: σ = 32M/πd³, τ = 16T/πd³, von Mises σ′ = √(σ² + 3τ²).

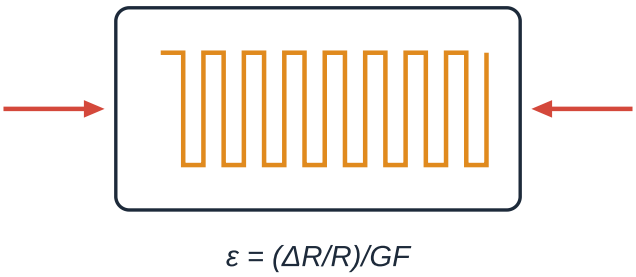
Diagrams



Mohr's circle for σx = 12, σy = −4, τxy = 6 ksi.



Shaft under bending plus torsion: the surface element sees σ and τ.



$$\varepsilon = (\Delta R/R)/GF$$

Strain-gage rosette reading at a point.

2. Essential formulas

Name	Formula	Variables / units
Principal stresses	$\sigma_{1,2} = (\sigma x+\sigma y)/2 \pm \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Max in-plane shear	$\tau_max = \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x-\sigma y)$	
Combined axial + bending	$\sigma = P/A \pm Mc/I$	
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$	

3. Handbook search keywords

Mohr's circle | principal stresses | combined stresses | von Mises

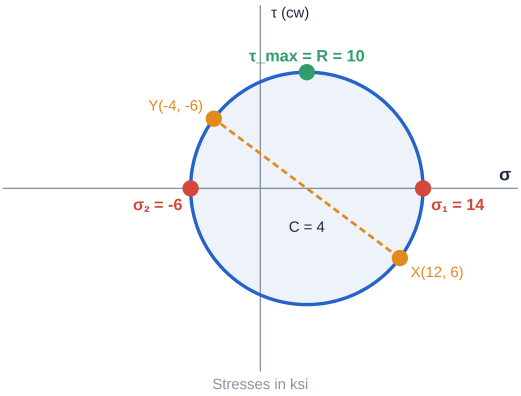
4. Worked examples

Example 1: Principal stresses
σx = 12 ksi, σy = −4 ksi, τxy = 6 ksi. Find σ_1, σ_2 and τ_max.
Handbook search: **Mohr's circle**

1. C = 4 ksi; R = √(8² + 6²) = 10 ksi.
2. σ_1 = 14 ksi; σ_2 = -6 ksi; τ_max = 10 ksi.

Answer: σ_1 = 14, σ_2 = −6, τ_max = 10 ksi

Check your answer: $\sigma_1 + \sigma_2$ must equal σx + σy (8 ksi). That is a quick invariant check.

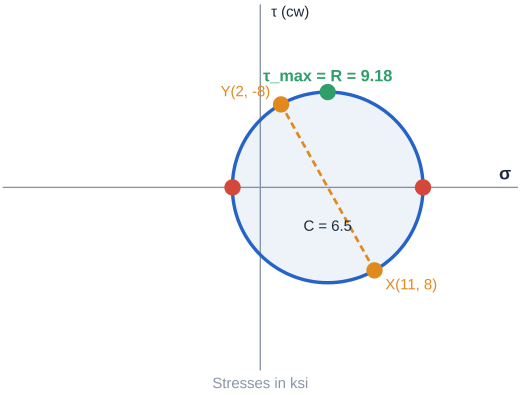


Try it yourself: Max in-plane shear
σx = 11 ksi, σy = 2 ksi, τxy = 8 ksi. Find τ_max (in-plane).
Handbook search: **maximum shear stress**

1. C = (11 + 2)/2 = 6.5 ksi.
2. R = √(4.5² + 8²) = 9.18 ksi = τ_max.

Answer: 9,179 psi

Check your answer: τ_max = R, and it must be at least |τxy| (8 ksi).

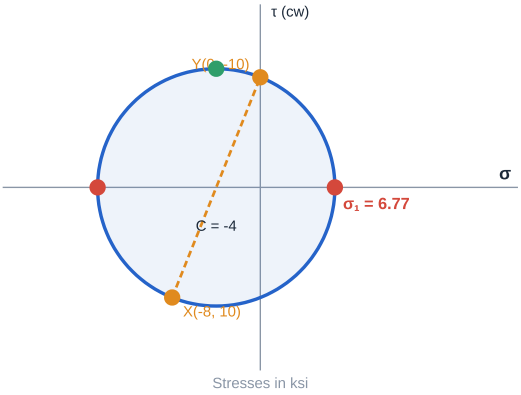


Example 3: Negative normal stress
σx = −8 ksi, σy = 0, τxy = 10 ksi. Find σ_1 and σ_2.
Handbook search: **principal stresses**

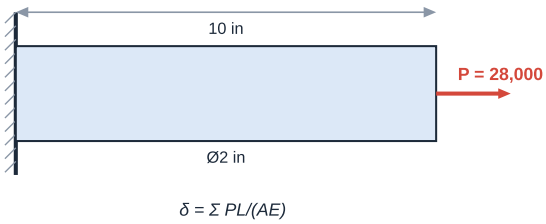
1. C = −4 ksi; R = √(4² + 10²) = 10.77 ksi.
2. σ_1 = C + R = 6.77 ksi; σ_2 = C − R = -14.77 ksi.

Answer: σ_1 = 6.77 ksi; σ_2 = -14.77 ksi

Check your answer: $\sigma_1 + \sigma_2$ = −8 ksi = σx + σy. ✔



Example 4: Axial plus bending
A 2 × 4 in bar carries 28,000 lb tension and 22,000 in-lb about its strong axis. Find σ_max.
Handbook search: **combined stress**

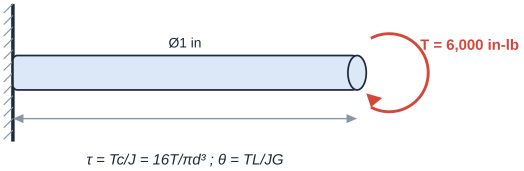


1. σ_axial = P/A = 28,000/8 = 3,500 psi.
2. S = bh²/6 = 2 × 4²/6 = 5.333 in³; σ_b = M/S = 4,125 psi.
3. σ_max = 7,625 psi (tension side).

Answer: 7,625 psi

Check your answer: On the other face the stress is -625 psi. The two faces should differ by 2σ_b.

Example 5: Shaft: bending + torsion
Ø1.0 in shaft with M = 6,000 in-lb and T = 6,000 in-lb. Find the von Mises stress.
Handbook search: **von Mises**

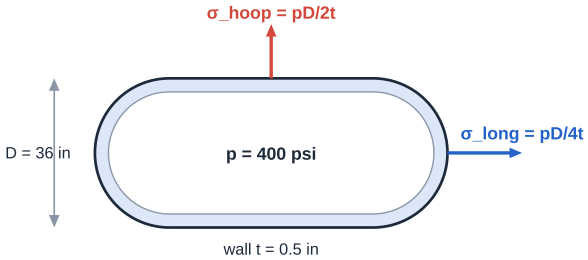


1. σ = 32M/πd³ = 61,115 psi; τ = 16T/πd³ = 30,558 psi.
2. σ' = √(σ² + 3τ²) = 80,848 psi.

Answer: 80,848 psi

Check your answer: σ' must exceed σ alone (61,115 psi) because τ adds to it.

Example 6: Thin-wall vessel stresses
A tank of 36 in mean diameter with a 0.5 in wall holds 400 psi. Find the hoop and longitudinal stresses.
Handbook search: **thin wall pressure vessel**



1. σ_h = pD/(2t) = 400 × 36/1 = 14,400 psi.
2. σ_L = pD/(4t) = 7,200 psi.

Answer: σ_h = 14,400 psi; σ_L = 7,200 psi

Check your answer: Hoop stress is always twice the longitudinal stress.

5. Common traps

- Forgetting $\sigma_3 = 0$ when σ_1 and σ_2 have the same sign (then τ_abs = $\sigma_1/2$).
- Sign errors for compressive σy.
- Using √(σ² + 4τ²) (Tresca) when von Mises is asked.

2.F Combined Loading: one-page cheat summary

- ► $C = (\sigma x + \sigma y)/2$
- ► $R = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau^2]}$
- ► $\sigma_{1,2} = C \pm R$
- ► $\tau_{max} = R$ (in-plane)
- ► $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Formulas

Principal stresses	$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Max in-plane shear	$\tau_{max} = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x - \sigma y)$
Combined axial + bending	$\sigma = P/A \pm Mc/I$
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Search keywords

Mohr's circle, principal stresses, combined stresses, von Mises

Top traps

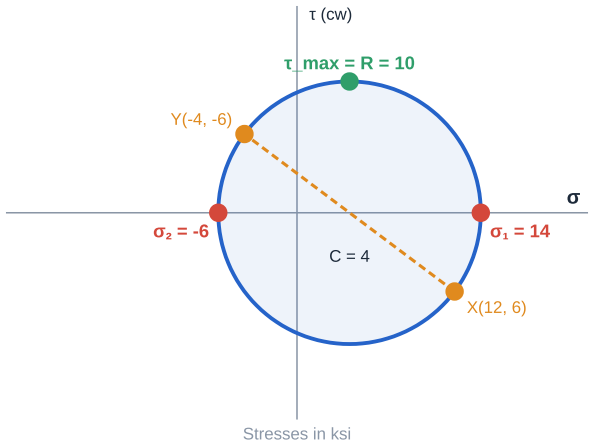
- Forgetting $\sigma_3 = 0$ when σ_1 and σ_2 have the same sign (then $\tau_{abs} = \sigma_1/2$).
- Sign errors for compressive σy.
- Using $\sqrt{(\sigma^2 + 4\tau^2)}$ (Tresca) when von Mises is asked.

2.G Static Failure

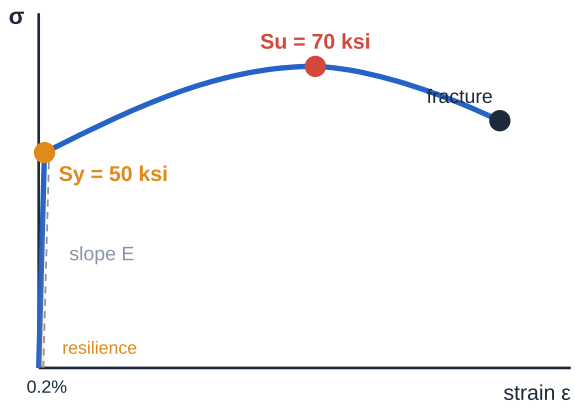
1. Key concepts

- Ductile materials: use distortion energy (von Mises, the best fit) or maximum shear stress (Tresca, conservative). Brittle materials: use maximum normal stress, Coulomb–Mohr or modified Mohr.
- von Mises plane stress: $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} = \sqrt{(\sigma x^2 - \sigma x \sigma y + \sigma y^2 + 3\tau^2)}$. n = Sy/σ′.
- Tresca: n = Sy/(σ_max − σ_min), including $\sigma_3 = 0$. Shear yield: von Mises Ssy = 0.577Sy; Tresca 0.5Sy.
- Static stress concentration: ductile materials can usually ignore Kt (local yielding relieves it). Brittle materials must apply Kt.

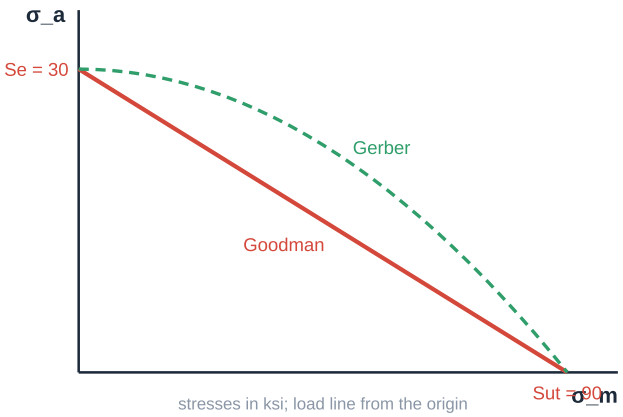
Diagrams



Principal stresses feeding the failure theories.



Ductile material: yield governs static failure.



Failure envelopes are drawn on stress axes.

2. Essential formulas

Name	Formula	Variables / units
von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$	
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$	
Shear yield	$Ssy = 0.577Sy\ (DE), 0.5Sy\ (MSS)$	
Max normal (brittle)	$n = Sut/\sigma_1$	

3. Handbook search keywords

distortion energy theory | maximum shear stress theory | factor of safety | Mohr theory brittle

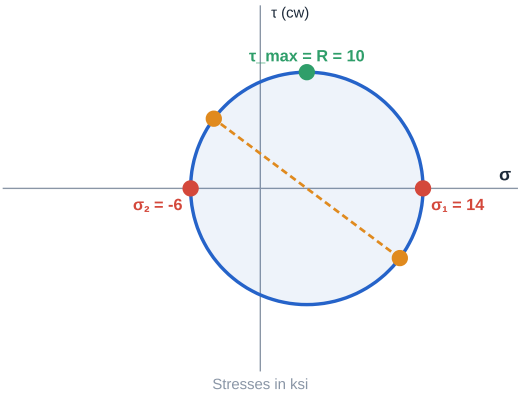
4. Worked examples

Example 1: von Mises vs Tresca
Use $\sigma_1 = 14$ ksi and $\sigma_2 = -6$ ksi from the 2.F example, with Sy = 50 ksi. Find n by both theories.
Handbook search: **distortion energy**

1. $\sigma' = \sqrt{(14^2 - 14(-6) + 6^2)} = 17.78$ ksi → n = 2.81.
2. Tresca: $\sigma_1 - \sigma_3 = 14 - (-6) = 20$ ksi → n = 2.50.

Answer: n_DE = 2.81; n_Tresca = 2.50

Check your answer: Tresca is always the same as or more conservative than von Mises.

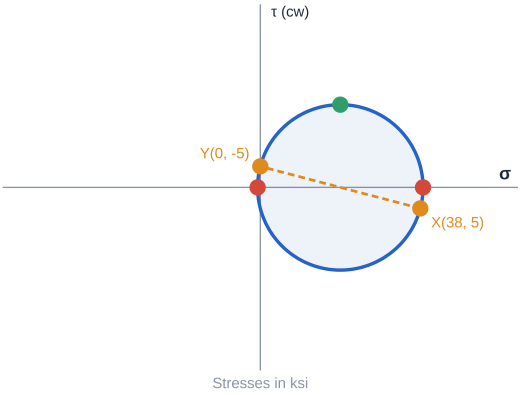


Try it yourself: Von Mises factor of safety
σx = 38,000 psi, τxy = 5,000 psi, Sy = 60,000 psi. Find n (distortion energy).
Handbook search: **von Mises**

1. $\sigma' = \sqrt{(\sigma x^2 + 3\tau^2)} = 38{,}974$ psi.
2. n = Sy/σ' = 1.54.

Answer: n = 1.54

Check your answer: With small τ, σ' is close to σx, so n should be close to 60/38 = 1.58.

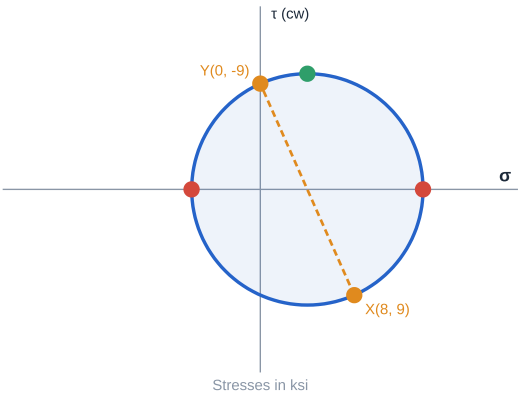


Example 3: Tresca factor of safety
Sy = 50 ksi; σx = 8,000 psi, σy = 0, τxy = 9,000 psi. Find n by Tresca.
Handbook search: **maximum shear stress theory**

1. $\sigma_1 = 13{,}849$ psi; $\sigma_2 = -5{,}849$ psi (opposite signs).
2. $\tau_{max} = (\sigma_1 - \sigma_2)/2 = R = 9{,}849$ psi.
3. n = (Sy/2)/τ_max = 25,000/9,849 = 2.54.

Answer: n = 2.54

Check your answer: When σ_1 and σ_2 have opposite signs, use $\sigma_1 - \sigma_2$, not σ_1 alone.

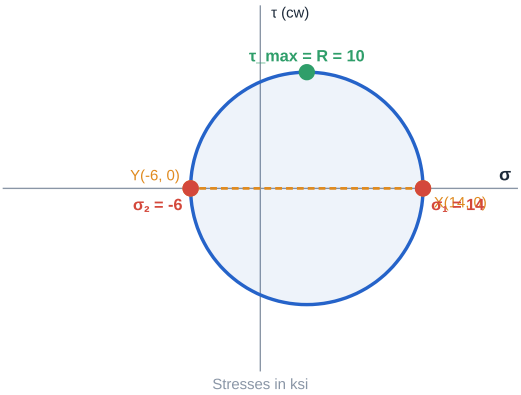


Example 4: Both theories, same point
$\sigma_1 = 14$ ksi, $\sigma_2 = -6$ ksi, Sy = 50 ksi. Compare n by von Mises and Tresca.
Handbook search: **distortion energy**

1. $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} = 17.78$ ksi → n = 2.81.
2. Tresca: $\sigma_1 - \sigma_2 = 20$ ksi → n = 50/20 = 2.50.

Answer: n_vM = 2.81; n_Tresca = 2.50

Check your answer: Tresca (2.50) ≤ von Mises (2.81). ✔

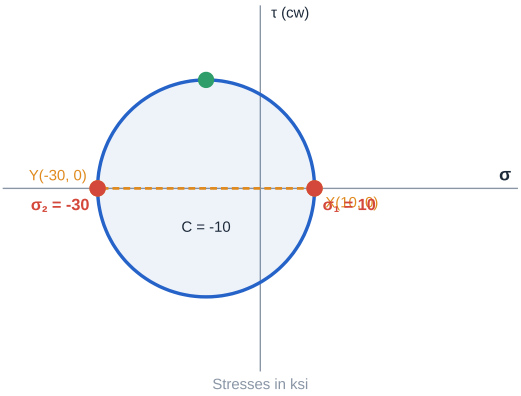


Example 5: Brittle material (modified Mohr / Coulomb-Mohr)
Cast iron: Sut = 40 ksi, Suc = 120 ksi. $\sigma_1 = 10$ ksi, $\sigma_2 = -30$ ksi. Find n by Coulomb–Mohr.
Handbook search: **Coulomb-Mohr**

1. $1/n = \sigma_1/Sut - \sigma_2/Suc = 10/40 + 30/120 = 0.500$.
2. n = 2.00.

Answer: n = 2.00

Check your answer: Brittle parts use Sut and Suc, not Sy. n should be less than 40/10 = 4.



5. Common traps

- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.G Static Failure: one-page cheat summary

- ► $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$
- ► Tresca $n = Sy/(\sigma max - \sigma min)$
- ► Ssy: $0.577Sy$ / $0.5Sy$
- ► Brittle: Coulomb–Mohr
- ► Ductile static: ignore Kt

Formulas

von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$
Shear yield	$Ssy = 0.577Sy$ (DE), $0.5Sy$ (MSS)
Max normal (brittle)	$n = Sut/\sigma_1$

Search keywords

distortion energy theory, maximum shear stress theory, factor of safety, Mohr theory brittle

Top traps

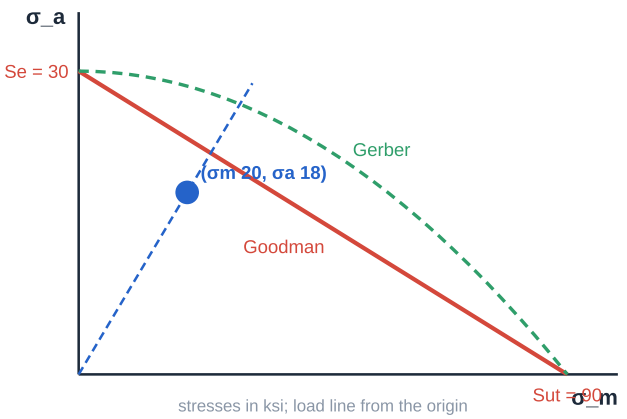
- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.H Fatigue Failure

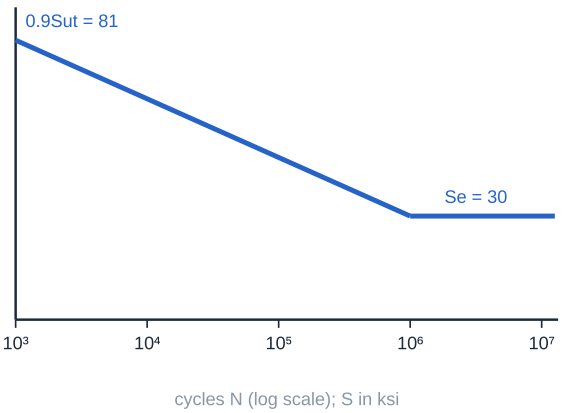
1. Key concepts

- The endurance limit of steel: Se′ ≈ 0.5Sut (Sut ≤ 200 ksi), or 100 ksi above that. Correct it with the Marin factors Se = ka·kb·kc·kd·ke·Se′ (surface, size, load, temperature, reliability).
- Fluctuating stress: σa = (σmax − σmin)/2, σm = (σmax + σmin)/2. Apply K_f = 1 + q(Kt − 1) to the stresses.
- Mean-stress criteria: modified Goodman 1/n = σa/Se + σm/Sut (conservative, the usual exam choice). Gerber is a parabola (better fit). ASME-elliptic uses Sy. Also check first-cycle yield: σa + σm ≤ Sy/n.
- Finite life: S-N line through (10^3, f·Sut) and (10^6, Se), with S = aN^b. Miner's rule: $\Sigma n_i/N_i = 1$.

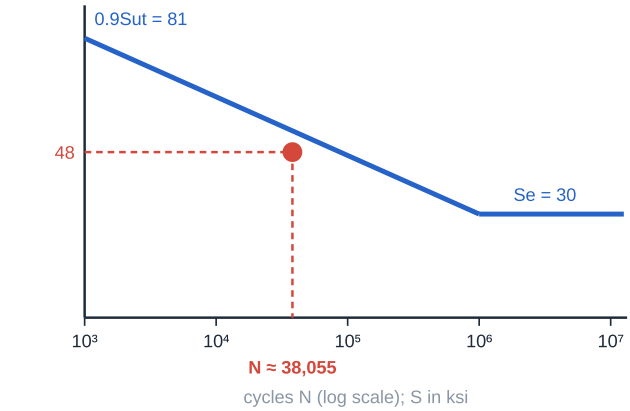
Diagrams



Modified Goodman line with a load line from the origin.



S-N curve: 0.9Sut at 10^3 cycles down to Se at 10^6.



Finite life at σa = 48 ksi.

2. Essential formulas

Name	Formula	Variables / units
Endurance limit	Se′ = 0.5Sut (steel)	
Surface factor	ka = a·Sut^b (machined: a = 2.70, b = −0.265 ksi)	
Size factor	kb = 0.879d^−0.107 (0.11–2 in)	
Fatigue Kt	K_f = 1 + q(Kt − 1)	
Goodman	1/n = σa/Se + σm/Sut	

Basquin	$S_f = aN^b$; $b = -(1/3)\log(f_{Sut}/S_e)$; $a = (f_{Sut})^2/S_e$	
Miner	$\Sigma\, n_i/N_i = 1$	

3. Handbook search keywords

endurance limit · Marin factors · modified Goodman · notch sensitivity · S-N diagram · Miner's rule

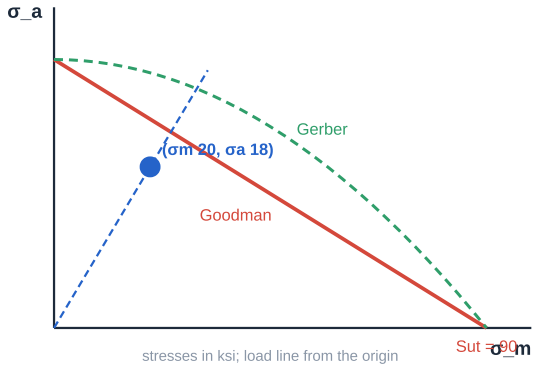
4. Worked examples

Example 1: Goodman factor of safety
A machined Ø1.0 in steel shaft (Sut = 90 ksi) sees σa = 18 ksi and σm = 20 ksi (K_f already applied), in rotating bending. Find n by Goodman.
Handbook search: **modified Goodman**

1. ka = 2.70(90)^−0.265 = 0.819; kb = 0.879(1.0)^−0.107 = 0.879.
2. Se = 0.5(90)(0.819)(0.879) = 32.4 ksi.
3. 1/n = 18/32.4 + 20/90 → n = 1.29.

Answer: n ≈ 1.29

Check your answer: The operating point (σm, σa) must sit inside the Goodman line when n > 1.



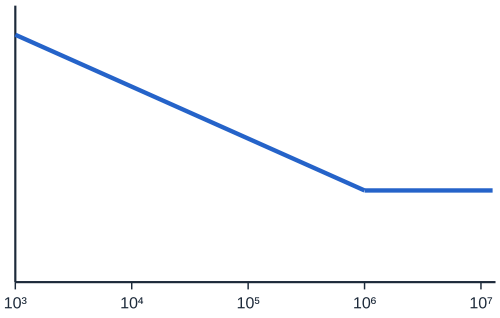
stresses in ksi; load line from the origin

Example 2: Fatigue stress concentration
Kt = 2.4 and q = 0.85. Find K_f.
Handbook search: **fatigue stress concentration**

1. K_f = 1 + 0.85(1.4) = 2.19.

Answer: 2.19

Check your answer: K_f always falls between 1 and K_t.



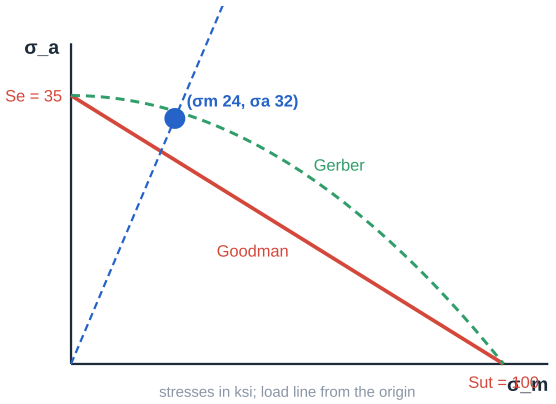
cycles N (log scale); S in ksi

Try it yourself: Goodman (n < 1)
Su = 100 ksi, Se = 35 ksi, σa = 32 ksi, σm = 24 ksi. Find n (modified Goodman).
Handbook search: **modified Goodman**

1. 1/n = σa/Se + σm/Su = 32/35 + 24/100 = 1.1543.
2. n = 0.866.

Answer: n = 0.866

Check your answer: n < 1 predicts finite life. The point sits outside the line on the diagram.



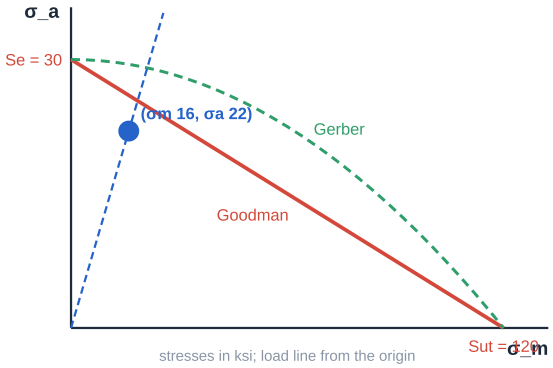
stresses in ksi; load line from the origin

Example 4: Gerber criterion
Sut = 120 ksi, Se = 30 ksi, σa = 22 ksi, σm = 16 ksi. Find n (Gerber).
Handbook search: **Gerber**

1. n = ½(Sut/σm)²(σa/Se)[−1 + √(1 + (2σmSe/(Sut σa))²)].
2. n = 1.321.

Answer: n = 1.321

Check your answer: Gerber is less conservative than Goodman (Goodman gives 1.154).



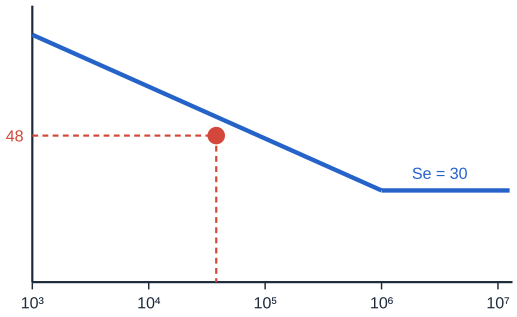
stresses in ksi; load line from the origin

Example 5: Finite life from the S-N line
Sut = 90 ksi, Se = 30 ksi, reversed σa = 48 ksi. Using Sf = aN^b through (10³, 0.9Sut) and (10⁶, Se), find N.
Handbook search: **S-N diagram**

1. a = (0.9Sut)²/Se = 218.7 ksi; b = −(1/3)log(0.9Sut/Se) = -0.1438.
2. N = (σa/a)^(1/b) = 38,055 cycles.

Answer: ≈ 38,055 cycles

Check your answer: 48 ksi lies between 30 and 81 ksi, so N must fall between 10³ and 10⁶.



cycles N (log scale); S in ksi

5. Common traps

- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.
- Skipping the first-cycle yield check.

2.H Fatigue Failure: one-page cheat summary

- ▶ $S_e' = 0.5S_{ut}$
- ▶ $S_e = k_a k_b k_c k_d k_e S_e'$
- ▶ $K_f = 1 + q(K_t - 1)$
- ▶ $\sigma_a = (\max - \min)/2$; $\sigma_m = (\max + \min)/2$
- ▶ Goodman: $1/n = \sigma_a/S_e + \sigma_m/S_{ut}$
- ▶ Miner: $\Sigma n/N = 1$

Formulas

Endurance limit	$S_e' = 0.5S_{ut}$ (steel)
Surface factor	$k_a = a S_{ut}^b$ (machined: a = 2.70, b = −0.265 ksi)
Size factor	$k_b = 0.879 d^{-0.107}$ (0.11–2 in)
Fatigue Kt	$K_f = 1 + q(K_t - 1)$
Goodman	$1/n = \sigma_a/S_e + \sigma_m/S_{ut}$
Basquin	$S_f = aN^b$; $b = -(1/3)\log(f S_{ut}/S_e)$; $a = (f S_{ut})^2/S_e$
Miner	$\Sigma\, n_i/N_i = 1$

Search keywords

endurance limit, Marin factors, modified Goodman, notch sensitivity, S-N diagram, Miner's rule

Top traps

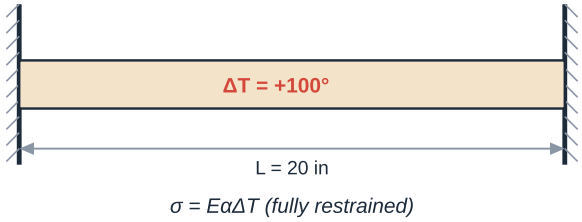
- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.

2.I Thermal and Interference Stresses

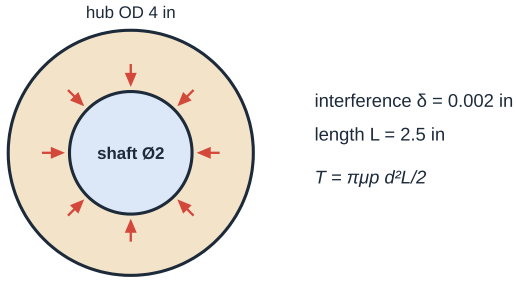
1. Key concepts

- A fully restrained bar heated by ΔT: σ = EαΔT (compressive). If there is a gap, only the expansion beyond the gap is restrained.
- Press and shrink fits: the interface pressure p comes from the interference δ (diametral). Same material, solid shaft: p = Eδ(D_o² − d²)/(2dD_o²).
- Hub hoop stress at the bore is σ_t = p(D_o² + d²)/(D_o² − d²). Torque capacity T = πμpd²L/2. Axial force capacity F = πμpdL.
- Shrink-fit assembly: the hub temperature rise is ΔT = (δ + clearance)/(αd).

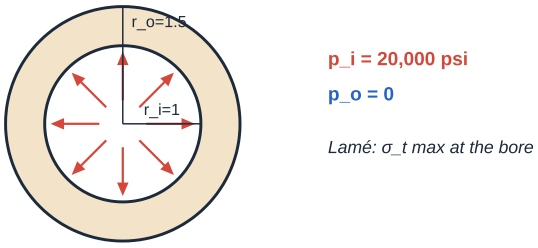
Diagrams



Restrained bar: σ = EαΔT.



Press fit: interference creates contact pressure p.



Thick cylinder: Lamé stresses.

2. Essential formulas

Name	Formula	Variables / units
Restrained bar	$\sigma = E\alpha\Delta T$	with gap: σ = E(αΔTL − gap)/L
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$	same E, solid shaft
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$	
Torque capacity	$T = \pi\mu p d^2 L/2$	
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$	

3. Handbook search keywords

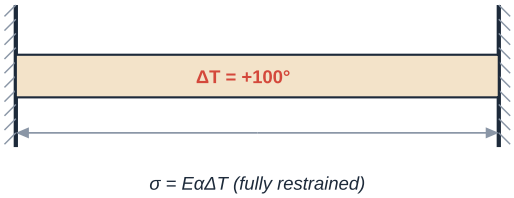
thermal stress · interference fit pressure · shrink fit · press fit torque

4. Worked examples

Example 1: Restrained rail
A steel bar ($E = 29\times10^6$ psi, $\alpha = 6.5\times10^{-6}$/°F) is fixed between walls and heated 100 °F. Find σ.
Handbook search: **thermal stress**

1. $\sigma = E\alpha\Delta T = 29\times10^6(6.5\times10^{-6})(100) = 18{,}850$ psi (compression).

Answer: 18,850 psi



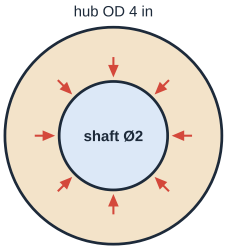
σ = EαΔT (fully restrained)

Check your answer: σ does not depend on length when the bar is fully restrained.

Example 2: Press-fit torque
A Ø2.0 in steel shaft is pressed into a steel hub (OD 4.0 in, L = 2.5 in) with $\delta = 0.002$ in, $E = 30\times10^6$ psi, $\mu = 0.15$. Find p and T.
Handbook search: **interference fit**

1. $p = 30\times10^6(0.002)(16-4)/(2\cdot2\cdot16) = 11{,}250$ psi.
2. $T = \pi(0.15)p(2^2)(2.5)/2 = 26{,}507$ in-lb.

Answer: T ≈ 26,507 in-lb



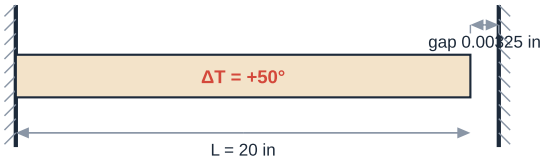
interference δ = 0.002 in
length L = 2.5 in
T = πμp d²L/2

Check your answer: Torque capacity grows with p, μ, length and d².

Try it yourself: Thermal stress with a gap
Bar: $E = 29\times10^6$ psi, $\alpha = 6.5\times10^{-6}$/°F, L = 20 in, gap 0.00325 in, heated 50 °F. Find σ.
Handbook search: **thermal stress**

1. Free growth = $\alpha\Delta TL = 0.0065$ in.
2. Blocked growth = 0.0065 − 0.00325 = 0.00325 in.
3. $\sigma = E \times$ blocked/L = 4,712 psi (compression).

Answer: 4,712 psi



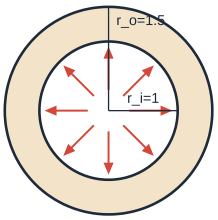
σ = E(αΔTL − gap)/L once the gap closes

Check your answer: If the free growth were smaller than the gap, σ would be zero.

Example 4: Thick cylinder, internal pressure
$r_i = 1$ in, $r_o = 1.5$ in, $p_i = 20{,}000$ psi. Find the maximum hoop stress.
Handbook search: **Lamé thick wall**

1. $\sigma_{t,max}$ (at r_i) $= p(r_o^2 + r_i^2)/(r_o^2 - r_i^2) = 52{,}000$ psi.

Answer: 52,000 psi



p_i = 20,000 psi
p_o = 0
Lamé: σ_t max at the bore

Check your answer: The thin-wall estimate pr/t = 40,000–60,000 psi brackets it, and Lamé is higher at the bore.

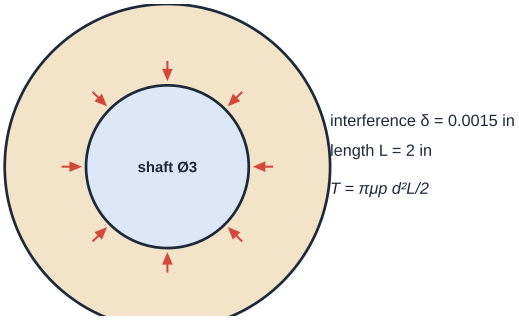
Example 5: Shrink-fit heating
A steel hub ($\alpha = 6.5\times10^{-6}$/°F, 3 in bore) must clear 0.0015 in interference plus 0.001 in assembly clearance. Find ΔT.

Handbook search: **shrink fit**

1. Required growth = 0.0025 in = $\alpha D \Delta T$.
2. $\Delta T = 0.0025/(6.5\times10^{-6} \times 3) = 128$ °F.

Answer: 128 °F

Check your answer: A few hundred °F is typical. Keep it below the tempering temperature of the hub.



5. Common traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.

2.I Thermal and Interference Stresses: one-page cheat summary

- ► $\sigma = E\alpha\Delta T$
- ► $p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
- ► $T = \pi\mu p d^2 L/2$
- ► $\Delta T_{shrink} = (\delta + c)/(\alpha d)$

Formulas

Restrained bar	$\sigma = E\alpha\Delta T$
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$
Torque capacity	$T = \pi\mu p d^2 L/2$
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$

Search keywords

thermal stress, interference fit pressure, shrink fit, press fit torque

Top traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.